Data-Driven Nests

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Demand Estimation

Discrete Choice Models

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- When shocks $\sim$ Type I EV $\implies$ Multinomial logit
  - Closed form solutions of choice probability + low number of parameters
  - Generates unrealistic substitution patterns (IIA) Blue-bus/Red-bus
- A common alternative is Nested Logit (NL)
  - Natural extension of Multinomial Logit

Nested demand models are ubiquitous across many fields

- Industrial Organization: Verboven (1996), Parry & Small (2005), Einav (2007), Grennan (2013), Ciliberto & Williams (2014), Berry et al. (2016), Björnerstedt & Verboven (2016), Allcott et al. (2024)...

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$\Longrightarrow$ In many applications, demand can be appropriately described by a nesting structure

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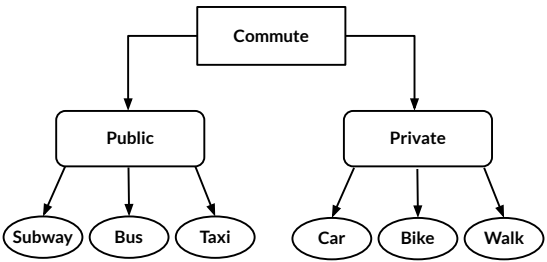

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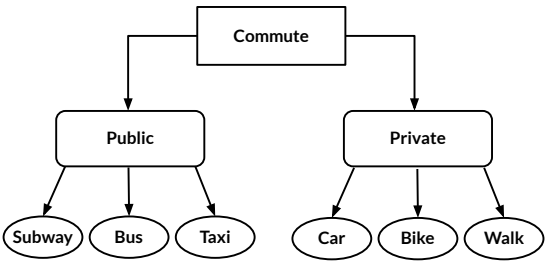

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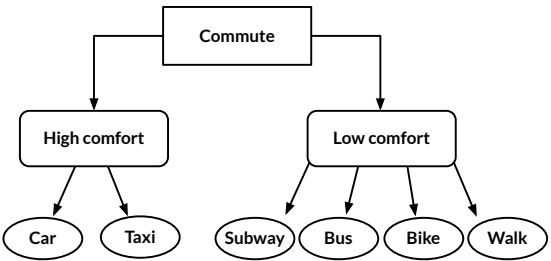

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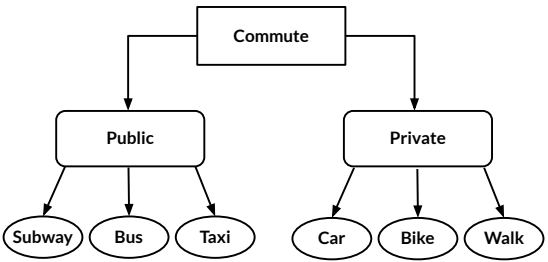

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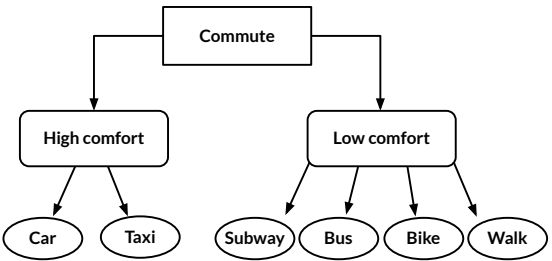

$\Longrightarrow$ Mis-specification of nests can lead to biased estimates (Fosgerau et al., 2024)

This paper: Estimating nests in discrete choice models

- Identification relies on **exploiting the structure of the model**

- Two-step estimation procedure:

  1. Cluster products based on **model-implied demand curves** (Bonhomme and Manresa, 2015)
     → Demand curves share coefficients **within nests** but differ **across nests**
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- Empirical application: Demand for beer
  - In this setting, data-driven NL outperforms Naive Nested Logit and Mixed Logit out of sample

Related Literature

- **Discrete choice models of random utility with nests:** McFadden (1978, 1981), Berry (1994), Verboven (1996), Cardell (1997), McFadden and Train (2000), Grigolon and Verboven (2014), Fosgerau, Monardo & De Palma (2024)
- **Group fixed effect estimator:** Han & Moon (2010), Bonhomme & Manresa (2015), Choi and Okui (2024)
- **Eliciting grouping structure:** Kovach & Tserenjigmid (2022), Hortacsu, Lieber, Monardo & de Paula (ongoing)

Our contribution: Recover the nesting structure from ***aggregate demand data*** for settings with ***many products*** exploiting the structure of the model

Outline

1. Empirical model
2. Practical aspects and extensions
3. Consistency, statistical properties, and Monte Carlo
4. Empirical application: Demand for beer

Empirical model

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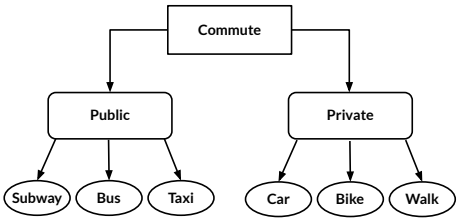

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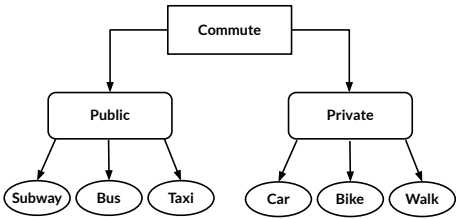
– Nest structure ~ common within-nest shock
  $\rightarrow \varepsilon_{ijm} = \eta_{i,k(j),m} + \tilde{\varepsilon}_{ijm}$ (Cardell, 1997)



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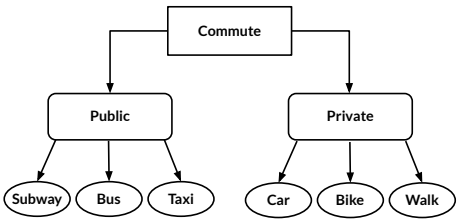
  $\rightarrow$ if $\sigma^{k} = 0 \implies \eta_{k(j),m}$ dominates
  $\rightarrow$ if $\sigma^{k} = 1 \implies$ back to logit



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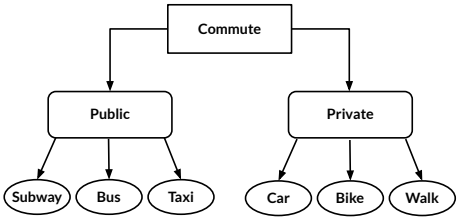
- Nests lead to different degrees of substitution:
  → More substitution **within** than **across** nests



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With a slight abuse of notation for ξ_{jm}, it follows:

$$\log \mathbb{P}_{jm} = \frac{\beta_x x_{jm} + \beta_p p_{jm} + \xi_{jm}}{\sigma^{k(j)}} + \lambda_m^{k(j)} = \underbrace{\beta_x^{k(j)}}_{\beta_x/\sigma^k} x_{jm} + \underbrace{\beta_p^{k(j)}}_{\beta_p/\sigma^k} p_{jm} + \lambda_m^{k(j)} + \xi_{jm}.$$

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Grouping on model-implied demand curves

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  $\rightarrow$ With different σ^k, marginal effect of covariates differ across nests $\implies$ different elasticities
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Identification requires:

- **Group separation:** Cannot have $\beta^k = \beta^{k'}$ AND $\lambda_m^k = \lambda_m^{k'}$, $\forall m$ for some $k \neq k'$
- **Non-nested grouping structure in error:** $Corr(\xi_{jm}, e(j)) \neq 0$ such that $k(j) \not\subseteq e(j)$

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Simultaneously search over:

1. Coefficients: $(\beta^1, \ldots, \beta^K, \lambda^1, \ldots, \lambda^K)$

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For simplicity, let's consider $\mathbb{E}[\xi_{jm} | x_{jm}, p_{jm}, \lambda_m^k] = 0$, and take means across markets:

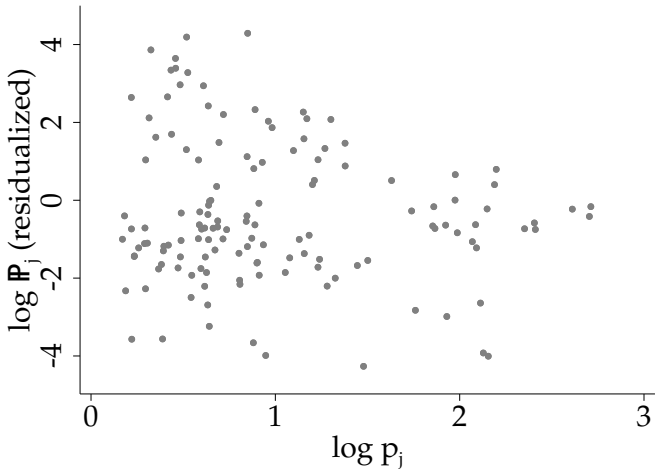
$$\overline{\log \mathbb{P}_j} = \beta_x^{k(j)} \bar{x}_j + \beta_p^{k(j)} \overline{p_j} + \overline{\lambda}^{k(j)}, \text{ where } \overline{\xi_j} \to 0$$

Key observation: There are K lines with coefficients $(\beta_x^{k(j)}, \beta_p^{k(j)}, \overline{\lambda}^{k(j)})$ that perfectly fit the data

Simultaneously search over:

1. Coefficients: $(\beta^1, \ldots, \beta^K, \lambda^1, \ldots, \lambda^K)$
2. Data partitions: $\{k(j)\}_{j=1}^J$ to assign products to different lines

Intuition on nest identification: graphic example with $K = 2$



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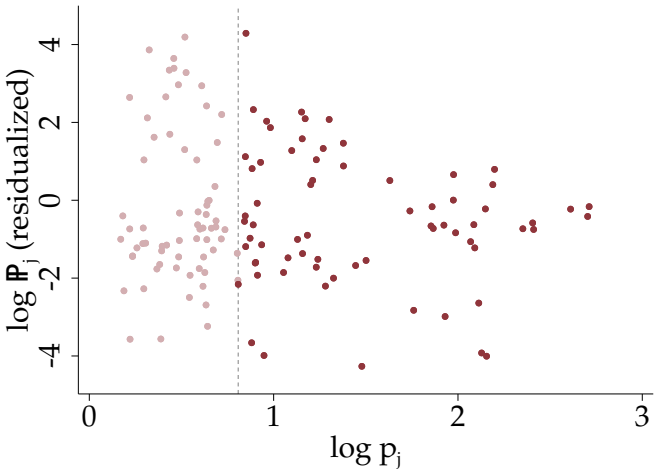

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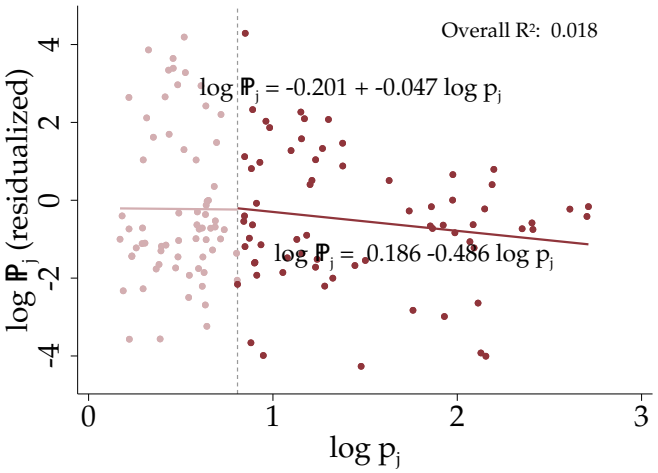

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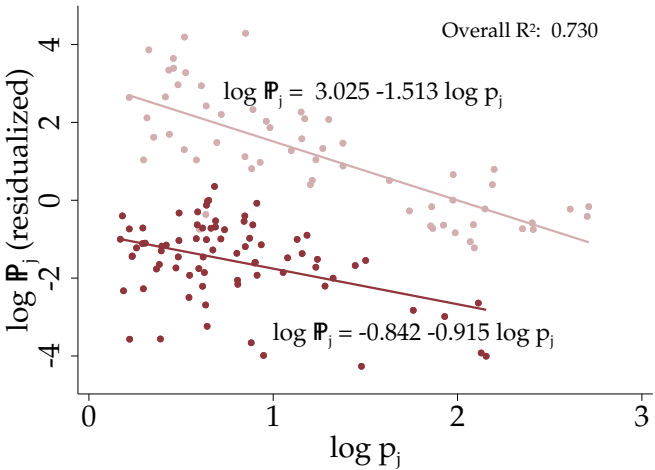

Outline

1. Empirical model
   - 1.1 Identification
   - 1.2 Estimation

2. Practical aspects and extensions

3. Consistency, statistical properties, and Monte Carlo

4. Empirical application: Demand for beer

Estimation of groups

First-step: Classification

Group-fixed effect estimator defined by the following optimization problem:

$$\underset{\substack{k(1),\ldots,k(J)\\ \beta^1,\ldots,\beta^K,\lambda_1^1,\ldots,\lambda_M^K}}{\arg\min} \sum_{m=1}^{M}\sum_{j=1}^{J}\left(\log \mathbb{P}_{jm} - (\beta_p^{k(j)} p_{jm} + \beta_x^{k(j)} x_{jm} + \lambda_m^{k(j)})\right)^2$$

Extends k-means by minimizing distance to regression lines (instead of cluster centroids):

- If groups were known, this is just a regression with group interactions
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Combinatorial, non-convex problem!
→ Approach: iterative clustering algorithm extending Bonhomme and Manresa (2015)

Two-step strategy

Second step: Estimation of structural parameters

Once we have grouping structure $\{k(1), \ldots, k(J)\}$, estimate β and σ^k as if groups were known

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We follow Berry (1994):

$$\log \frac{\mathbb{P}_{jm}}{\mathbb{P}_{0m}} = \beta_p p_{jm} + \beta_x x_{jm} + (\sigma^{k(j)} - 1) \log \mathbb{P}_{j,m|k(j)} + \xi_{jm}$$

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Address (potential) endogeneity:

- Instrument for $\log \mathbb{P}_{j,m|k(j)} = \frac{e^{\frac{\beta_p p_{jm} + \beta_x x_{jm}}{\sigma^{k(j)}}}}{\sum_{d \in B_{k(j)}} e^{\frac{\beta_p p_{jm} + \beta_x x_{jm}}{\sigma^{k(j)}}}}$ $\implies$ Functions of $x_{j'm}$ for $j' \in k(j)$
- Instrument for prices, z_{jm}
  $\rightarrow$ Note: Restrict to settings with relevant instruments as $J \rightarrow \infty$, Amstrong (2016)

Practical aspects and extensions

Outline

1. Empirical model

2. Practical aspects and extensions
   - 2.1 Endogenous prices
   - 2.2 Choosing the number of groups
   - 2.3 Introducing consumer heterogeneity
   - 2.4 Higher-level nesting structures
   - 2.5 Type II EV error: Nested CES

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4. Empirical application: Demand for beer

Endogenous prices with an excluded shifter

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$\Longrightarrow$ Restricted to a "2SLS" strategy

Grouped fixed effects 2SLS strategy

1. First predict prices on x_{jm} and z_{jm} to get $\hat{p}_{jm}$
2. The classify on projected prices $\hat{p}_{jm}$ instead of p_{jm}:

$$\log \mathbb{P}_{jm} = \beta_x^{k(j)} x_{jm} + \beta_p^{k(j)} \hat{p}_{jm} + \underbrace{\nu_{jm}}_{\xi_{jm} + \beta_p^{k(j)} \hat{v}_{jm}} + \lambda_m^{k(j)},$$

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$\implies$ Robust approach is to also elicit price groups $k'(j)$, where potentially $k' \neq k$:

$$\underset{\substack{k'(1),\ldots,k'(J) \\ \gamma^1,\ldots,\gamma^{K'},\kappa_1^1,\ldots,\kappa_M^{K'}}}{\arg\min} \sum_{m=1}^{M} \sum_{j=1}^{J} \left(p_{jm} - (\gamma_x^{k'(j)} x_{jm} + \gamma_z^{k'(j)} z_{jm} + \kappa_m^{k'(j)}) \right)^2$$

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Note: Groupings in prices arise naturally in many common settings
→ e.g. Multiproduct firms with common marginal costs across product portfolio

Discussion

We **cannot** accommodate:

- Unobservables with grouping structure $e(j)$ that cannot be absorbed by true groups, $k(j) \subseteq e(j)$
- Non-linear utility in parameters
  → No theory has been successfully developed for non-linear models (yet?)
  → Tackling endogeneity would require an alternative strategy, like a control function approach
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- J growing large
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- M growing large (but slower than J):
  - In the presence of ξ_{jm}, this ensures probability of misclassification $\to 0$ as $M \to \infty$ Details

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2. Practical aspects and extensions
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Choosing K: Cross-validation with Elbow method

We can estimate the number of groups using a **cross-validation** procedure

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For $k \in \{1, \ldots, K\}$:

- Divide products into N equal subsets, $P_1, \ldots, P_N$.
- Pick subset P_n and estimate grouping structure and grouping parameters in the other $N - 1$ parts.
- Classify products across estimated groups in part P_n and **compute out-of-sample MSE**

$$MSE_n(k) = \frac{1}{J \cdot M} \sum_{m=1}^{M} \sum_{j \in P_n} (y_j - \beta_{m,-n}^{k(j)} x_j - \lambda_{m,-n}^{k(j)})^2$$

- Take average across N folds:

$$MSE(k) = \frac{1}{N} \sum_{n=1}^{N} MSE_n(k)$$

- Choose k according to

$$k^* = \arg\min_k \{MSE(k) \mid \text{marginal improvements become small}\}.$$

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Individual heterogeneity with observed conditional shares

Assume heterogeneity can be described by types t (e.g. income quintiles) and utility given by:

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$$\log \mathbb{P}_{jm}^{t} = \beta^{k^{t}(j)} x_{jm} + \xi_{jm} + (\sigma_{k^{t}(j)} - 1) \log IV_{k^{t}(j),m}^{t} - \log IV_{m}^{t}$$

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Two cases:

1. No common nesting structures: classify even type-by-type
2. Common nesting structures: solve joint problem

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Higher-level nesting structures

Upstream and downstream nests given by $A_1, \ldots, A_N$ and $B_1, \ldots, B_K$, respectively.

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so that

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We can run same classification algorithm!

Note: needs normalization of some $\lambda_m^k = 0$ to avoid co-linear group-market fixed effects.

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Type II EV error

A representative consumer maximizes a two-nested CES utility function over products j:

$$U_m = \left(\sum_{k=1}^{K} \left(\sum_{j \in B_k} \left(A_{jm}\, c_{jm} \right)^{(\sigma^k - 1)/\sigma^k} \right)^{\frac{\sigma^k}{\sigma^k - 1} \cdot \frac{\sigma^0 - 1}{\sigma^0}} \right)^{\sigma^0/(\sigma^0 - 1)},$$

where

- c_{jm} is consumption of product j,
- A_{jm} is a quality shifter,
- $\sigma^k > 1$ is the within-nest elasticity of substitution,
- $\sigma^0 \geq 1$ is the across-nest elasticity.

Type II EV error

Standard expenditure minimization gives the market share:

$$\mathbb{P}_{jm} = \frac{\left(A_{jm}/p_{jm}\right)^{\sigma^k-1}}{\sum_{j'\in B_k}\left(A_{j'm}/p_{j'm}\right)^{\sigma^k-1}} \cdot \frac{P_{km}^{1-\sigma^0}}{\sum_{k'=1}^{K} P_{k'm}^{1-\sigma^0}} = \left(\frac{A_{jm}}{p_{jm}}\right)^{\sigma^{k(j)}-1} \Lambda_m^{k(j)},$$

where $\Lambda_m^{k(j)}$ collects all nest- and market-level terms.

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where $\Lambda_m^{k(j)}$ collects all nest- and market-level terms.

Taking logs and setting $\log A_{jm} = \beta_x x_{jm} + \xi_{jm}$:

$$\log \mathbb{P}_{jm} = \underbrace{(\sigma^{k(j)} - 1)\beta_x}_{\beta_x^{k(j)}} x_{jm} - \underbrace{(\sigma^{k(j)} - 1)}_{-\beta_p^{k(j)}} \log p_{jm} + \lambda_m^{k(j)} + \xi_{jm},$$

which exactly maps to the first-step equation of our baseline model.

Consistency, statistical properties, and Monte Carlo

Regularity conditions with unknown nests

1. Let $J \to \infty$, $M \to \infty$, and $\frac{\log M}{J} \to 0$
2. K fixed and balanced nests: $|B_k| = O_p(J)$
3. Orthogonality of errors: $\mathbb{E}[\xi_{jm}, v_{jm} | x_{jm}, z_{jm}, \lambda_m^k] = 0$
4. Variation in x_{jm}, z_{jm} conditional on groups and thin tails

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We show asymptotic equivalence to the infeasible estimator:

- Convergence rate is $O_p(JM^{-\eta})$ for any $\eta > 0 \implies$ super-consistency
- Can estimate second step as if groups were known

Monte Carlo design: Data generating process

Fix $K = 3$. Set $\sigma = (0.2, 0.3, 0.6)$. Classify products using uniform draws $k(j) \sim \mathcal{U}\{1, ..., K\}$

Models for the average utility δ_{jm} and prices p_{jm} are given by:

$$\delta_{jm} = \beta_p p_{jm} + \beta_x x_{jm} + \xi_{jm} \quad \text{and} \quad p_{jm} = mc_{jm} + \rho \cdot \xi_{jm}$$

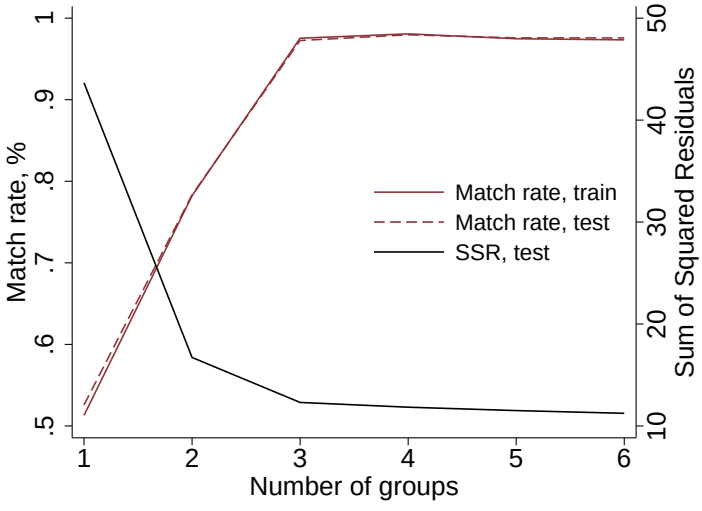
Generate data as follows:

$$\begin{bmatrix} \mu_x^k \\ \mu_{mc}^k \end{bmatrix} \overset{i.i.d.}{\sim} \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & 0.3 \\ 0.3 & 1 \end{bmatrix} \right) \implies \begin{bmatrix} x_{jm} \\ mc_{jm} \\ \xi_{jm} \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu_x^k \\ \mu_{mc}^k \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & 0.3 & 0 \\ 0.3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right)$$

Set $\rho = 0.3$

Estimating the number of groups: Monte Carlo

Set $K^0 = 3$, the total number of products $J = 500$, and the total number of markets is $M = 10$.



Results with $K = 3$

50 Bootstrap iterations

					β_p	β_c	σ_1	σ_2	σ_3
J	M	Runtime	Matched	True	-1	1	0.2	0.3	0.6
100	10	00:01	0.988	Mean β	-1.001	0.992	0.195	0.298	0.599
				Std β	0.037	0.031	0.013	0.007	0.01
100	50	00:18	1.0	Mean β	-1.0	1.003	0.2	0.3	0.6
				Std β	0.012	0.012	0.001	0.002	0.004
100	100	01:42	1.0	Mean β	-1.001	1.001	0.2	0.3	0.6
				Std β	0.008	0.008	0.001	0.001	0.003
500	10	00:14	0.995	Mean β	-1.0	0.998	0.199	0.3	0.6
				Std β	0.013	0.012	0.006	0.005	0.003
500	50	03:59	1.0	Mean β	-1.001	1.0	0.2	0.3	0.6
				Std β	0.006	0.004	0.001	0.001	0.001
500	100	17:42	1.0	Mean β	-1.0	1.0	0.2	0.3	0.6
				Std β	0.004	0.003	0.0	0.001	0.001
1000	10	00:15	0.995	Mean β	-1.001	1.002	0.2	0.3	0.599
				Std β	0.01	0.011	0.002	0.002	0.002
1000	50	08:08	1.0	Mean β	-1.002	1.001	0.2	0.3	0.6
				Std β	0.003	0.004	0.0	0.0	0.001
1000	100	31:48	1.0	Mean β	-1.0	1.0	0.2	0.3	0.6
				Std β	0.003	0.003	0.0	0.0	0.001

Notes: Running time for a 32-core machine with 16GB of RAM per core.

Empirical application: Demand for beer

Outline

1. Empirical model

2. Practical aspects and extensions

3. Consistency, statistical properties, and Monte Carlo

4. Empirical application: Demand for beer
   - 4.1 Data description and empirical model
   - 4.2 First-step estimation: Classification
   - 4.3 Out-of-sample group validation
   - 4.4 Second-step estimation
   - 4.5 Comparison with alternative empirical models

Nielsen IQ store panel data

Data Description:

- We define products as Universal Product Codes (UPCs)
- Focus on year 2022 and on UPCs categorized as beer (24,188 unique UPCs)
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- Each observation contains data on:
  - Total number of sales by UPC and state, aggregating across stores
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Nielsen IQ store panel data

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Keep products that are sold in at least ten states

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Keep products that are sold in at least ten states

- Unbalanced panel with $J = 2,806$ unique UPCs (11.6%) and 64,497 observations
- Cover between 61.5% to 99% of volume sales across states
- Average UPC is sold across 23.32 states $\sim$ 50% missing markets

Summary statistics

Large variation in shares and covariates

Panel A: Full Sample Summary Statistics

Variable	Mean	Std. Dev.	10th pct	Median	90th pct
Price ($)	10.618	6.758	2.596	10.298	20.308
Unit quantity (units)	8.838	8.229	1.000	6.000	24.000
Price per ml ($)	0.004	0.001	0.002	0.004	0.005
Unit size (ml)	535.627	450.385	354.882	354.882	739.338
Market share	0.859%	1.152%	0.061%	0.498%	1.967%

Panel B: Median Characteristics of the top ten national brands

Brand	Market share	Unit price/ml	Unit price	Unit quantity	Unit size
Bud Light	12.53%	0.003	8.63	6.00	473.18
Miller	7.81%	0.003	12.47	12.00	354.88
Modelo	7.63%	0.005	10.23	6.00	354.88
Coors	7.11%	0.003	12.62	12.00	354.88
Bud	6.73%	0.003	3.22	1.00	739.34
Michelob	5.96%	0.004	15.20	12.00	354.88
Corona	4.59%	0.004	16.52	12.00	354.88
Busch	4.43%	0.002	10.87	12.00	354.88
Natural	3.63%	0.002	6.49	6.00	473.18
New Belgium	1.98%	0.005	10.28	1.00	567.81

Empirical Model

We model the average utility δ_{jm} as follows:

$$\delta_{jm} = \beta_p \log \text{price}_{jm} + \beta_s \log \text{size}_j + \beta_q \log \text{quantity}_j + \eta_m + \xi_{jm}$$

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Instrument for prices: Gandhi-Houde instruments

$$\text{gh}^1_{jm} = \sum_{j' \notin \text{Brand}(j,m)} (\log \text{size}_j - \log \text{size}'_j)^2 \quad \text{and} \quad \text{gh}^2_{jm} = \sum_{j' \notin \text{Brand}(j,m)} (\log \text{quantity}_j - \log \text{quantity}'_j)^2$$

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We later consider an extension with categorical variables based on NielsenIQ beer types.

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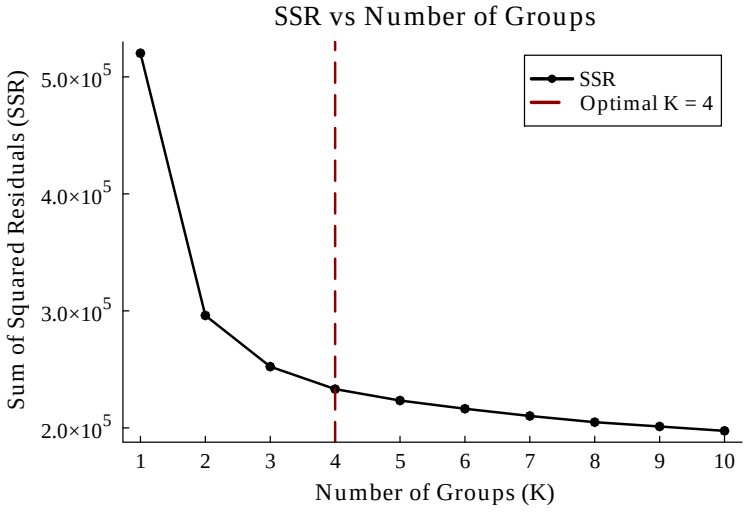
First-step: Choosing the number of groups

Elbow method suggest $K^* = 4$



First-step: Group characteristics

Characteristics	Unconditional values	Conditional values by group			
		1	2	3	4
Unit size (ml)	354.882	473.176	354.882	354.882	354.882
Unit quantity	6.000	6.000	6.000	6.000	6.000
Unit price	10.209	9.857	10.156	12.367	10.509
Price per ml	0.004	0.003	0.004	0.004	0.005
Share of domestic beer	0.794	0.820	0.773	0.830	0.861
Share of ale	0.158	0.225	0.098	0.291	0.377
Share of regular beer	0.442	0.469	0.430	0.442	0.326
Share of stout and porter	0.015	0.019	0.011	0.018	0.046
Share of light beer	0.369	0.266	0.446	0.238	0.239
Share of other types	0.016	0.022	0.015	0.010	0.012
Share of top ten brands	0.624	0.460	0.755	0.382	0.334
# of products	2806	659	242	1102	803
Label		Regular Lager	Light Beer	Special /Craft	Boutique Craft

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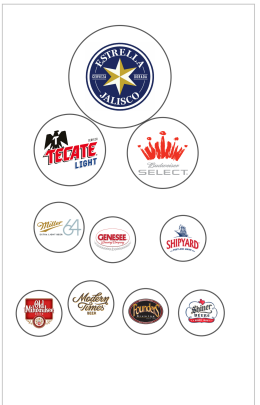
Brands by group

Group 1

Group 2

Group 3

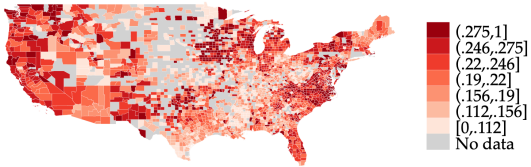


Group 4

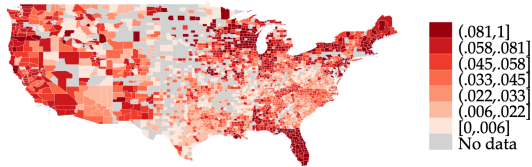
Logos are trademarks of their respective owners

Spatial correlation of county market shares by group

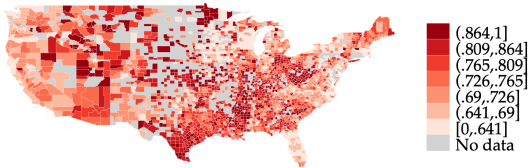
Regular lager



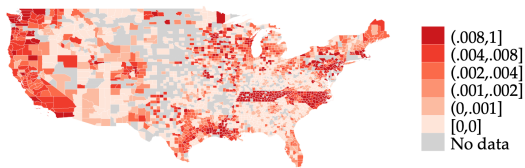
Special / craft



Light beer

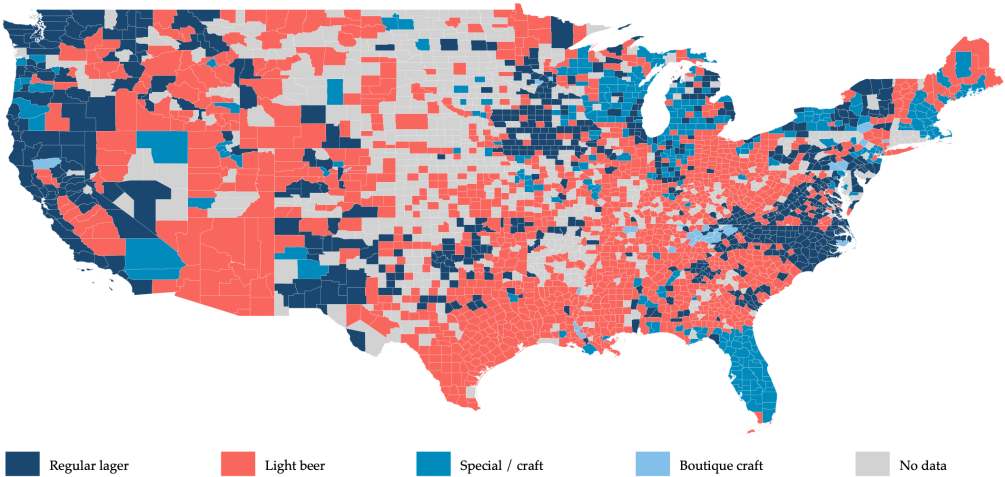


Boutique craft

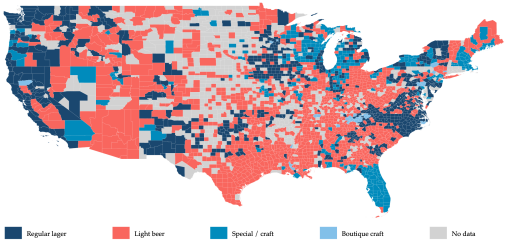


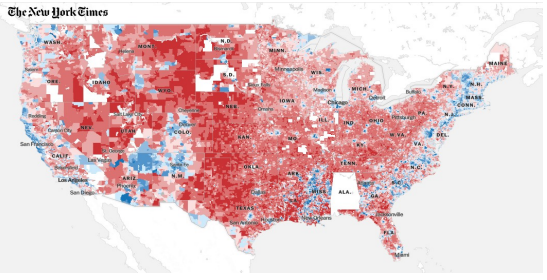
Most popular group (relative to national trends)

High degree of spatial correlation



Group popularity correlated with political outcomes!





Main takeaway: capturing unobserved drivers of demand

Relationship between alcohol preferences and political ideology has been established in the past
→ e.g., Yakovlev and Guessford (2013)

Alcohol Consumption and Political Ideology: What's Party Got to Do with It?*

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aawe
Journal of
Wine Economics
American Association of Wine Economists

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Main takeaway: The Data-driven NL can capture unobserved drivers of demand

- As long as they can be described (or well approximated) by a grouping structure
  - In our case, it's geography and/or political ideology:

$$\iota_m \quad \text{or} \quad \mathbb{1}\{m = \text{republican state}\}$$

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Group validation: Bud Light boycott

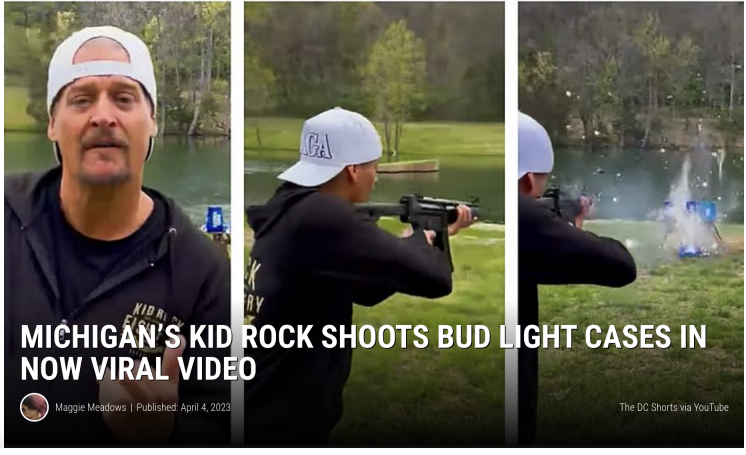
Testing out-of-sample nest substitution

In April 2023, Dylan Mulvaney—an actress, social media influencer, and transgender woman—releases a social media promotion of Bud Light

Group validation: Bud Light boycott

Testing out-of-sample nest substitution

The promotion sparked a backlash and calls for a boycott



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The boycott appeared to have some bite, and its effects were widely covered in the media

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Use Bud Light boycott as an exogenous negative shock to demand for Bud Light

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Hypothesis: if consumers are substituting consumption to other beers

$\Longrightarrow$ The boycott should lead to greater substitution inside "Light Beer" than toward other nests

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Test: Empirical model

$$\log \mathbb{P}_{jmt} = \kappa_{jm} + \eta_t + \sum_{k=1}^{K} \alpha_k \cdot \mathbb{1}\{\widehat{k}(j) = k\}\mathbb{1}\{t = 2023\} + \epsilon_{jmt}$$

by augmenting our state-product panel of sales by adding data for 2023

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by augmenting our state-product panel of sales by adding data for 2023

Because of UPC-market fixed effects κ_{jm}

$\Longrightarrow$ exploit within-product temporal variation not used in the classification of groups

Group validation: DiD results

Dependent Variable: Log Share								
	First stage		DiD				Placebo	
t=2023	-0.358***	(0.005)	-0.402***	(0.005)	-0.444***	(0.008)	-0.355***	(0.010)
Log Prices	-0.234***	(0.054)	-0.244***	(0.054)	-0.259***	(0.054)	-0.217***	(0.054)
Treatment (Budlight)	**-0.155****	(0.026)						
Treatment same group			0.425***	(0.008)				
Regular beer × t=2023					0.177***	(0.012)		
Light beer × t=2023					0.468***	(0.010)		
Non-lager/craft × t=2023					0.002	(0.013)		
Boutique craft × t=2023					0.017	(0.020)		
Placebo Regular beer × t=2023							-0.020	(0.014)
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Placebo Boutique craft × t=2023							0.009	(0.015)
Observations	177784		172778		172778		172778	
UPC-Market FE	YES		YES		YES		YES	
Exclude Bud Light	No		YES		YES		YES	
Baseline group	**Non-Budlight**		Other groups		Outside option		Outside option	

Notes: Heteroskedasticity-robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

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Second-step results: Structural parameters and matrix of elasticities

Rich heterogeneity in substitution patterns

	OLS	IV
log unit size	0.070 (0.006)***	0.339 (0.025)***
log upc quantity	0.080 (0.005)***	0.441 (0.026)***
log price	-0.054 (0.006)***	-0.416 (0.033)***
$\sigma_{\text{regular/lager}}$	0.142 (0.001)***	0.523 (0.014)***
$\sigma_{\text{light beer}}$	0.343 (0.002)***	0.807 (0.016)***
$\sigma_{\text{special/craft}}$	-0.015 (0.001)***	0.336 (0.013)***
$\sigma_{\text{boutique craft}}$	-0.320 (0.001)***	0.042 (0.015)***
Mean Own-price Elasticity	1.351 (0.003)	-3.561 (0.013)
Mean Cross-price Elasticity	-0.003 (0.008)	0.011 (0.033)
Market Fixed Effects	✓	✓
IV Type		GH_{units}, GH_{size}
Number of Products	2806	
Number of Observations	64497	

Literature range of beer own-price elasticities is $[-6.3, -2.4]$

(Pinske and Slade, 2004; Slade, 2004; Romeo, 2016; Miller and Weinberg, 2017)

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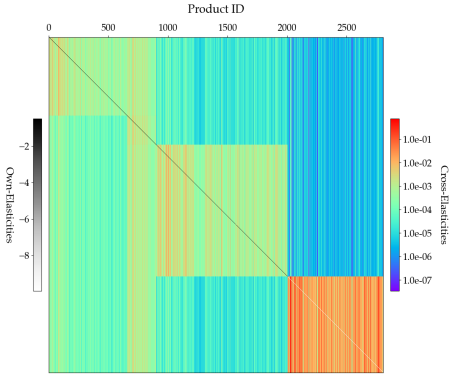

Literature range of beer own-price elasticities is $[-6.3, -2.4]$

(Pinske and Slade, 2004; Slade, 2004; Romeo, 2016; Miller and Weinberg, 2017)

Outline

1. Empirical model

2. Practical aspects and extensions

3. Consistency, statistical properties, and Monte Carlo

4. Empirical application: Demand for beer
   - 4.1 Data description and empirical model
   - 4.2 First-step estimation: Classification
   - 4.3 Out-of-sample group validation
   - 4.4 Second-step estimation
   - 4.5 Comparison with alternative empirical models

Naive Nested Logit: Using NielsenIQ "Beer Type" as Nests

	OLS	IV
log unit size	0.054 (0.005)***	0.686 (0.041)***
log upc quantity	0.059 (0.004)***	0.694 (0.042)***
log price	-0.039 (0.005)***	-0.753 (0.050)***
σ_{ale}	-0.029 (0.001)***	0.239 (0.012)***
$\sigma_{\text{light beer}}$	0.087 (0.001)***	0.348 (0.010)***
σ_{other}	-0.356 (0.002)***	-0.098 (0.015)***
$\sigma_{\text{regular/lager}}$	0.080 (0.001)***	0.322 (0.010)***
$\sigma_{\text{stout/porter}}$	-0.296 (0.001)***	0.040 (0.017)**
Mean Own-price Elasticity	0.360 (0.003)	-3.752 (0.104)
Mean Cross-price Elasticity	-0.001 (0.003)	0.007 (0.111)
Market Fixed Effects	✓	✓
IV Type		GH_{units}, GH_{size}
Number of Products	2806	
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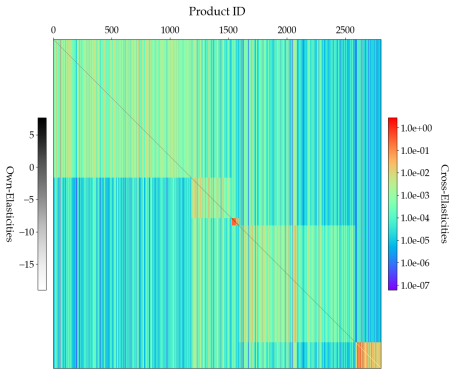

Bud Light boycott: Comparing Data-driven vs. Naive NL

Dependent Variable: Log Share										
	Data-driven NL				Naive NL				Interaction	
t=2023	-0.402***	(0.005)	-0.444***	(0.008)	-0.322***	(0.006)	-0.444***	(0.008)	-0.374***	(0.007)
Log Prices	-0.244***	(0.054)	-0.259***	(0.054)	-0.241***	(0.054)	-0.266***	(0.054)	-0.254***	(0.054)
Treatment same group	**0.425***	(0.008)								
Regular beer × t=2023			0.177***	(0.012)						
Light beer × t=2023			0.468***	(0.010)						
Non-lager/craft × t=2023			0.002	(0.013)						
Boutique craft × t=2023			0.017	(0.020)						
Naive light beer × t=2023					**0.131***	(0.016)	0.254***	(0.017)		
Naive ale × t=2023							0.029**	(0.012)		
Naive other × t=2023							0.013	(0.043)		
Naive lager × t=2023							0.254***	(0.012)		
Naive stout × t=2023							-0.040*	(0.021)		
DD Light only									0.387***	(0.009)
Naive Light only									0.033	(0.024)
DD and Naive Light									0.424***	(0.012)
Observations	172778		172778		172778		172778		172778	
UPC-Market FE	YES		YES		YES		YES		YES	
Exclude Bud Light	YES		YES		YES		YES		YES	
Baseline group	Other groups		Outside option		Outside option		Outside option		Outside option	

Notes: Heteroskedasticity-robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

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Mixed logit

We estimate the following mixed-logit model using PyBLP Conlon and Gortmaker (2020):

$$u_{ijm} = \delta_{jm} + \mu_{ijm} + \epsilon_{ijm},$$

where

$$\delta_{jm} = \beta_p \text{log price}_{jm} + \beta_s \text{log size}_j + \beta_q \text{log quantity}_j + \eta_m + \xi_{jm},$$

and μ_{ijm} is drawn from

$$\mu_{ijm} = \left(\sigma_p \nu_i\right) \text{log p}_{jm},$$

where σ_p governs the variance of the idiosyncratic component and $\nu_i \sim N(0, 1)$.

Mixed Logit Structural estimates

For this setting and specification, Mixed Logit looks close to a simple Logit

Panel A: Linear Coefficients	β
log price	-3.775*** (0.090)
log unit size	3.077*** (0.062)
log unit quantity	3.222*** (0.062)
Panel B: Non-linear Coefficient	σ_p
log price	0.002*** (0.0003)
Own-price elasticity	-4.683 (2.863)
Cross-price elasticity	0.002 (0.005)
Market Fixed Effects	✓
IV Type	Gandhi-Houde
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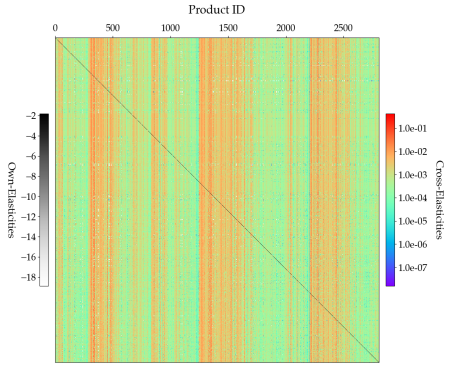

Mixed Logit

$$\log \mathbb{P}_{jmt} = \kappa_{jm} + \eta_t + f(\text{elasticity}^j_{BudLight})\mathbb{1}\{t = 2023\} + \epsilon_{jmt}$$

Dependent Variable: Log Share								
	Continuous		DiD					
t=2023	-0.289***	(0.008)	-0.290***	(0.008)	-0.312***	(0.006)	-0.449***	(0.008)
Log Prices	-0.163*	(0.084)	-0.240***	(0.054)				
$\mathcal{E}^i_{\text{Bud Light}} \times$ t=2023	**-7.048****	(1.585)						
Above median $\mathcal{E}^i_{\text{Bud Light}} \times$ t=2023			**-0.031****	(0.011)				
Top quartile $\mathcal{E}^i_{\text{Bud Light}} \times$ t=2023					**-0.003**	(0.012)		
Quartile 1 $\mathcal{E}^i_{\text{Bud Light}} \times$ t=2023							**0.219****	(0.014)
Quartile 2 $\mathcal{E}^i_{\text{Bud Light}} \times$ t=2023							0.085***	(0.013)
Quartile 3 $\mathcal{E}^i_{\text{Bud Light}} \times$ t=2023							0.107***	(0.014)
Quartile 4 $\mathcal{E}^i_{\text{Bud Light}} \times$ t=2023							**0.134****	(0.013)
Observations	107404		172778		172778		172778	
UPC-Market FE	YES		YES		YES		YES	
Exclude Bud Light	YES		YES		YES		YES	
Baseline group	Outside option		Below median		Other quartiles		Outside option	

Notes: Heteroskedasticity-robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Conclusions

Conclusions: Proposed a two-step estimator to estimate nesting structure

Showed conditions for consistency of estimator

Empirical application based on US beer market shows:

- Reasonable results and rich substitution patterns with **minimal data and computational requirements**
- Nests correlated with geography/political preferences
  → Data-driven Nested Logit can capture unobservable factors of demand
- For this application, our method outperforms Naive Nested Logit and Mixed Logit in one out-of-sample test

Thanks!

Appendix

Blue-bus/Red-bus problem

Implications of IIA property

A traveler has a choice of commuting by car or taking a blue bus

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Assume indirect utility from *car* and *blue bus* is the same so

$$P_{car} = P_{blue\ bus} = \frac{1}{2} \implies \frac{P_{car}}{P_{blue\ bus}} = 1$$

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Given IIA, we still have $\frac{P_{car}}{P_{blue\ bus}} = 1$. The only consistent model with both is

$$P_{car} = P_{blue\ bus} = P_{red\ bus} = \frac{1}{3}$$

Connection to IIA

$$\begin{aligned}\log \mathbb{P}_{jm} &= \frac{\beta_x x_{jm} + \beta_p p_{jm} + \xi_{jm}}{\sigma^{k(j)}} + \lambda_m^{k(j)} \\ &= \underbrace{\beta_x^{k(j)}}_{\frac{\beta_x}{\sigma^k}} x_{jm} + \underbrace{\beta_p^{k(j)}}_{\frac{\beta_p}{\sigma^k}} p_{jm} + \lambda_m^{k(j)} + \xi_{jm},\end{aligned}$$

Intuition:

- The marginal effect of variation in covariates varies by nest
- IIA within a nest: only j and j' covariates affect $\log \frac{\mathbb{P}_{jm}}{\mathbb{P}_{j'm}}$
- Not IIA across nests: If $k(j) \neq k(j')$, $\log \frac{\mathbb{P}_{jm}}{\mathbb{P}_{j'm}}$ also function of $\lambda_m^{k(j)}$ and $\lambda_m^{k(j')}$

One-step group estimation

With exogenous prices, combine steps 1 + 2 by solving the following constrained problem:

$$\underset{\substack{k(1),\ldots,k(J)\\ \beta^1,\ldots,\beta^K,\lambda_1^1,\ldots,\lambda_M^K\\ \beta,\sigma^1,\ldots,\sigma^K}}{\arg\min} \sum_{m=1}^{M}\sum_{j=1}^{J}\Big(\log\Big(\frac{\mathbb{P}_{jm}}{\mathbb{P}_{0m}}\Big)-(x_{jm}\beta^k+\lambda_m^k)\Big)^2,$$

where

$$\beta^k=\frac{\beta}{\sigma^k}\quad\text{and}\quad\lambda_m^k=(\sigma^k-1)\log\Big(\sum_{d\in B_k}e^{\log\left(\frac{\mathbb{P}_{dm}}{\mathbb{P}_{0m}}\right)-\lambda_m^k}\Big),$$

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but

- No theory has been developed for consistency of non-linear models
- In practice it may work, but substantial computational burden due to non-linearities
- Endogeneity of prices requires GMM estimation $\implies$ lack of identification Choi and Okui (2024)

Nash-Bertrand pricing

Assume marginal costs can be written as

$$\log mc_{jm} = \gamma_x^{c(j)} x_{jm} + \gamma_z^{c(j)} z_{jm} + \alpha_m^{c(j)} + \omega_{jm}$$

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Under Bertrand–Nash competition with single-product firms:

$$p_{jm} = \frac{1}{1 - \frac{1}{\mathcal{E}^j_{jm}}} \, mc_{jm}$$

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Substituting into prices:

$$\log p_{jm} = \gamma_x^{c(j)} x_{jm} + \gamma_z^{c(j)} z_{jm} + \underbrace{\left(\alpha_m^{c(j)} - \log\left(1 - \frac{\sigma_{k(j)}}{\beta}\right)\right)}_{\kappa_m^{k'(j)}} + \underbrace{\omega_{jm}}_{\nu_{jm}} + O_p(\log J^{-1})$$

Note: applied to many other pricing models and any general ownership structure

Classification algorithm

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4. Repeat until convergence of parameters.
5. In practice, to find global optimum, repeat for different starting values and take minimum

Regularity conditions with unknown nests

Details on regularity conditions to ensure consistent classification:

- $\lambda_m^{k(j)}$ is a random variable that diverges when $J \to \infty$. Conditions so they are well behaved:
  - Balanced nests: $|B_k| = O_p(J)$ for $k = 1, \dots K$
  - Normalization by $\log J$ to stay in compact space + conditions so that $\log J \lambda_m^k(J)$ is well-defined
  - Sequence $(\sigma_{01,J}, \dots, \sigma_{0K,J})_{J=1}^{\infty}$ such that $\lambda_m^{k(j)}$ is defined when J grows
- Group separation: cannot have $\beta^k = \beta^{k'}$ and $\lambda_m^k = \lambda_m^{k'}$ for all m for some $k \neq k'$

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- Group separation: cannot have $\beta^k = \beta^{k'}$ and $\lambda_m^k = \lambda_m^{k'}$ for all m for some $k \neq k'$
  - This requires that two groups contain same products and look *identical* across all markets

$$x_{jm} = x_j,\ z_{jm} = z_j\ \forall m$$

  - Either a very knife-edge GDP or for most common GDPs it happens with probability zero

GMM Cannot Jointly Recover Groups and Parameters

The GMM estimator solves:

$$\min_{\theta,\, k(\cdot)} \; \bar{m}(\theta, k(\cdot))^\top W \, \bar{m}(\theta, k(\cdot)), \qquad \bar{m}(\theta, k(\cdot)) = \frac{1}{JM} \sum_{j=1}^{J} \sum_{m=1}^{M} \begin{pmatrix} \xi_{jm}\left(\theta_{k(j)}\right) z_{jm} \\ \xi_{jm}\left(\theta_{k(j)}\right) x_{jm} \end{pmatrix}.$$

In the just-identified case, for *any* candidate $k(\cdot)$, one can *always* find $\hat{\theta}(k(\cdot))$ such that:

$$\bar{m}\left(\hat{\theta}(k(\cdot)),\ k(\cdot)\right) = 0$$

$\Longrightarrow$ the objective is zero for every $k(\cdot)$

$\Longrightarrow$ $k(.)$ and parameters cannot be jointly identified via GMM

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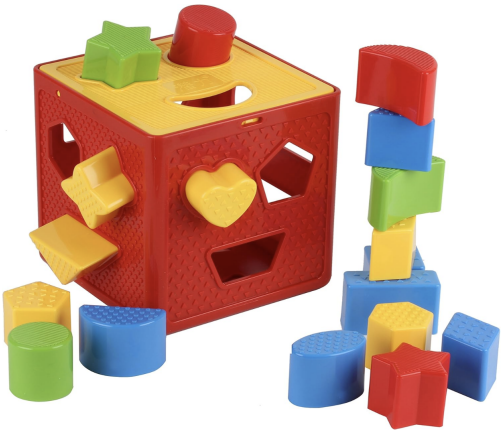
Result extends to over-identified case: for any grouping $k(\cdot)$, there exist parameters θ such that the GMM objective converges to zero in probability. Groups remain unidentified.

Why asymptotics on M?

Graphical (literal) toy example: playing in the dark



- You can't see
- Different shapes are placed in different urns
- Every minute m, you can learn by touching an unknown shape from an urn for 5 seconds
- Your goal is to get as many shapes through the box as possible
- The more minutes m, the more you learn!

Why asymptotics on M?

Simplified example

Back

- Consider the following simplified model with $K = 2$:

$$y_{jm} = \alpha^*_{k_j^*} + \xi_{jm}, \quad k_j \in \{1, 2\}.$$

- We characterize the misclassification probability:

$$\Pr\left(\widehat{k_j}(\alpha) = 2 \middle| k_j^* = 1\right) = \Pr\left(\left(\overline{y}_j - \alpha_2\right)^2 < \left(\overline{y}_j - \alpha_1\right)^2 \middle| k_j^* = 1\right).$$

- If ξ_{jm} are iid normal $(0, \sigma^2)$ and $\alpha_1 < \alpha_2$ then this is:

$$\Pr\left(\overline{\xi}_j > \frac{\alpha_1 + \alpha_2}{2} - \alpha_1^*\right) \quad = \quad 1 - \Phi\left(\frac{\sqrt{M}}{\sigma}\left(\frac{\alpha_1 + \alpha_2}{2} - \alpha_1^*\right)\right),$$

  which vanishes exponentially fast as M increases.

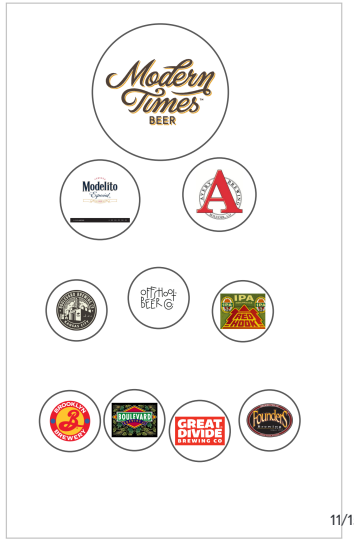
- **Intuition:** When M grows
  - Mean $\bar{y}_j$ converges to α_1
  - If mis-classified, then error $\bar{y}_j - \alpha_2$ eventually should become larger than $\bar{y}_j - \alpha_1$
  - Every m is a chance to learn: With enough opportunities, we'll eventually learn the truth!

Brands by group $k = 3$

Group 1

Group 2

Group 3



Brands by group $k = 5$

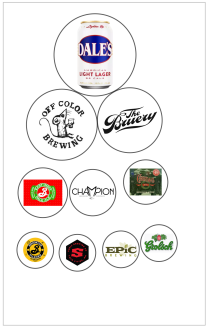
Group 1

Group 2

Group 3

Group 4

Group 5



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Classification including “Beer Type” categories

Table: Estimation results for $k^* = 3$

	OLS	IV
log unit size	0.037 (0.006)***	0.088 (0.024)***
log upc quantity	0.040 (0.005)***	0.122 (0.024)***
log price	-0.040 (0.006)***	-0.146 (0.031)***
σ_1	0.046 (0.001)***	0.303 (0.009)***
σ_2	0.269 (0.001)***	0.578 (0.010)***
σ_3	-0.209 (0.001)***	0.031 (0.009)***
Mean Own-price Elasticity	-0.276 (0.001)	-2.343 (0.006)
Mean Cross-price Elasticity	0.000 (0.002)	0.005 (0.015)
Market Fixed Effects	✓	✓
IV Type		GH_{units}, GH_{size}
Number of Products	2806	
Number of Observations	64497	

Pricing counterfactual: procedure

Plug DDNL estimates into a standard Bertrand–Nash counterfactual

Take $(\hat{\beta}_p, \{\hat{\sigma}_k\})$ from k^*=4 into the Bertrand–Nash FOC and simulate an ownership counterfactual

Multi-product FOC:

$$p_m = \mathrm{mc}_m - \Big[\underbrace{\Omega_m}_{\substack{\text{ownership} \\ \text{matrix}}} \circ \underbrace{\Delta_m(p_m; \hat{\theta}_D)}_{\substack{\text{share} \\ \text{Jacobian}}} \Big]^{-1} s_m(p_m; \hat{\theta}_D)$$

Procedure:

1. Recover $\widehat{\mathrm{mc}}_m$ from FOC under *observed* Ω_m^{obs}
2. Build Ω_m^{cf} (split joint owner of Firms A and B into two firms)
3. Solve FOC as a fixed point in p_m

Pricing counterfactual: Robustness on scenarios and K

Table: $\kappa = 0.264$, $e_B = 0.14$, $e_A = 0.10$.

		Case		Firm C		Firm A		Firm B	
Scenario	**Col.**	**Prof. wt.**	**Cost chg.**	Bench.	Ours	Bench.	Ours	Bench.	Ours
Baseline	(i)	yes	yes	+6.2	+4.6	+8.8	+4.0	+9.5	+1.3
No cost change	(ii)	yes	no	+7.3	+5.5	+13.5	+12.5	+15.9	+14.0
No profit weight	(iii)	no	yes	−0.5	−0.8	+1.1	−3.7	+3.2	−6.4
Ownership only	(iv)	no	no	+1.3	+0.8	+6.3	+4.6	+9.9	+6.2

Table: $\kappa = 0.374$ and varying the DDNL k^*

	$K = 3$	**K⋆ 4**	$K = 5$
$\hat{\beta}_p$	−1.038	**−0.416**	−0.084
Firm A $\Delta \log p$	−8.3%	**7.9%**	+4.4%
Firm B $\Delta \log p$	−12.9%	**5.3%**	−8.9%

Back